

codata

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NAME

codata – library for fundamental physical constants

LIBRARY

Codata library (*libcodata*, *-lcodata*)

SYNOPSIS

Fortran **use codata**

C **#include "codata.h"**

Python **import pycodata**

DESCRIPTION

codata is a Fortran library providing the fundamental physical constants published by CODATA. It includes the 2022, 2018, 2014 and 2010 recommended values, allowing applications to reproduce calculations based on specific CODATA releases.

The constants are implemented as a derived type that carries: *the name, the value, the uncertainty, and the unit*. All the constants are provided as **double precision reals**. Long constant names can be aliased with shorter names.

It provides:

- CODATA 2022 constants
- CODATA 2018 constants
- CODATA 2014 constants
- CODATA 2010 constants
- Native Fortran API
- C interoperability API
- Python bindings

DOCUMENTATION

The API is written as POSIX-style man pages which are distributed with the binaries.

HTML <https://milanskocic.github.io/codata/index.html>

PDF <https://milanskocic.github.io/codata/man/codata-bookman.pdf>

NOTES

The codata constants from the 2022 adjustment were integrated into the Fortran standard library (stdlib). This library is complementary to the constants defined in the stdlib by providing older values for the constants. See <https://github.com/fortran-lang/stdlib/releases/tag/v0.7.0>. If your application only requires the latest CODATA values, the Fortran standard library may be sufficient. Use codata when you need access to historical CODATA releases or compatibility with older scientific results.

To use codata within your fpm(1) (<https://github.com/fortran-lang/fpm>) project, add the following lines to your file:

```
[dependencies]
codata = { git="https://github.com/MilanSkocic/codata.git" }
```

If you only need sources for the CODATA constants that can be integrated directly into your project, you may be interested in the following repository: https://github.com/vmagnin/fundamental_constants.

LICENSE

MIT

SEE ALSO

codata(3), codata_cli(1), codata_version(3), codata_constant_type(3)

NAME

codata - library for fundamental physical constants

LIBRARY

codata (**-libcodata**, **-lcodata**)

SYNOPSIS

Fortran: *use codata*

C: *include "codata.h"*

Python: *import pycodata*

DESCRIPTION

codata(3) is a Fortran library providing the fundamental physical constants published by the CODATA.

Fortran API

exposes the constants as derived type `codata_constant_type` which carries:

- name
- value
- uncertainty
- unit

C API exposes a structure `codata_constant_type` that defines the same members as in Fortran:

- name
- value
- uncertainty
- unit

Python exposes a dictionary with the keys being:

- the name
- the value
- the uncertainty
- the unit

NOTES

The latest values (2022) do not have the year as a suffix in their name whereas older values (2010, 2014, 2018) feature the year as a suffix. All constants are declared with the same variable name in Fortran, C, and Python.

EXAMPLE

Example in Fortran

```
program example_in_f
  use codata
  implicit none
  print '(A)', '##### EXAMPLE IN FORTRAN #####'
  print '(A)', '# VERSION'
  print *, "version = ", version()
  print '(A)', '# CONSTANTS'
  print *, "c = ", SPEED_OF_LIGHT_IN_VACUUM%value
  print '(A)', '# UNCERTAINTY'
  print *, "u(c) = ", SPEED_OF_LIGHT_IN_VACUUM%uncertainty
  print '(A)', '# OLDER VALUES'
```

```

print '(A, F23.16)', "Mu_2022(latest) = ", MOLAR_MASS_CONSTANT%value
print '(A, F23.16)', "Mu_2018 = ", MOLAR_MASS_CONSTANT_2018%value
print '(A, F23.16)', "Mu_2014 = ", MOLAR_MASS_CONSTANT_2014%value
print '(A, F23.16)', "Mu_2010 = ", MOLAR_MASS_CONSTANT_2010%value
end program

```

Example in C

```

#include <stdio.h>
#include "codata.h"

int main(void){
    printf("%f", c);
    printf("##### EXAMPLE IN C #####0);
    printf("%s0, "# VERSION");
    printf("version = %s0, codata_version());
    printf("%s0, "# CONSTANTS");
    printf("c = %f0, SPEED_OF_LIGHT_IN_VACUUM.value);
    printf("%s0, "# UNCERTAINTY");
    printf("u(c) = %f0, SPEED_OF_LIGHT_IN_VACUUM.uncertainty);
    printf("%s0, "# OLDER VALUES");
    printf("Mu_2022(latest) = %23.16f0, MOLAR_MASS_CONSTANT.value);
    printf("Mu_2018 = %23.16f0, MOLAR_MASS_CONSTANT_2018.value);
    printf("Mu_2014 = %23.16f0, MOLAR_MASS_CONSTANT_2014.value);
    printf("Mu_2010 = %23.16f0, MOLAR_MASS_CONSTANT_2010.value);
    return 0;
}

```

Example in Python

```

import sys
sys.path.insert(0, "../py/src/")
import pycodata
print("##### EXAMPLE IN PYTHON #####")
print("# VERSION")
print(f"version = {pycodata.__version__}")
print("# Constants")
print("c =", pycodata.SPEED_OF_LIGHT_IN_VACUUM["value"])
print("# UNCERTAINTY")
print("u(c) =", pycodata.SPEED_OF_LIGHT_IN_VACUUM["uncertainty"])
print("# OLDER VALUES")
print("Mu_2022 =", pycodata.MOLAR_MASS_CONSTANT["value"])
print("Mu_2018 =", pycodata.MOLAR_MASS_CONSTANT_2018["value"])
print("Mu_2014 =", pycodata.MOLAR_MASS_CONSTANT_2014["value"])
print("Mu_2010 =", pycodata.MOLAR_MASS_CONSTANT_2010["value"])

```

SEE ALSO

[codata\(1\)](#), [codata_version\(3\)](#), [codata_constant_type\(3\)](#)

CODATA 2022

List of available constants:

- ALPHA_PARTICLE_ELECTRON_MASS_RATIO
- ALPHA_PARTICLE_MASS
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV
- ALPHA_PARTICLE_MASS_IN_U
- ALPHA_PARTICLE_MOLAR_MASS
- ALPHA_PARTICLE_PROTON_MASS_RATIO
- ALPHA_PARTICLE_RELATIVE_ATOMIC_MASS
- ALPHA_PARTICLE_RMS_CHARGE_RADIUS
- ANGSTROM_STAR
- ATOMIC_MASS_CONSTANT
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV
- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP
- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP
- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP
- ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP
- ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP
- ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP
- ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY
- ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY
- ATOMIC_UNIT_OF_ACTION
- ATOMIC_UNIT_OF_CHARGE
- ATOMIC_UNIT_OF_CHARGE_DENSITY
- ATOMIC_UNIT_OF_CURRENT
- ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM
- ATOMIC_UNIT_OF_ELECTRIC_FIELD
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT
- ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY
- ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL
- ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM
- ATOMIC_UNIT_OF_ENERGY
- ATOMIC_UNIT_OF_FORCE
- ATOMIC_UNIT_OF_LENGTH
- ATOMIC_UNIT_OF_MAG_DIPOLE_MOM
- ATOMIC_UNIT_OF_MAG_FLUX_DENSITY
- ATOMIC_UNIT_OF_MAGNETIZABILITY

- ATOMIC_UNIT_OF_MASS
- ATOMIC_UNIT_OF_MOMENTUM
- ATOMIC_UNIT_OF_PERMITTIVITY
- ATOMIC_UNIT_OF_TIME
- ATOMIC_UNIT_OF_VELOCITY
- AVOGADRO_CONSTANT
- BOHR_MAGNETON
- BOHR_MAGNETON_IN_EV_T
- BOHR_MAGNETON_IN_HZ_T
- BOHR_MAGNETON_IN_INVERSE_METER_PER_TESLA
- BOHR_MAGNETON_IN_K_T
- BOHR_RADIUS
- BOLTZMANN_CONSTANT
- BOLTZMANN_CONSTANT_IN_EV_K
- BOLTZMANN_CONSTANT_IN_HZ_K
- BOLTZMANN_CONSTANT_IN_INVERSE_METER_PER_KELVIN
- CHARACTERISTIC_IMPEDANCE_OF_VACUUM
- CLASSICAL_ELECTRON_RADIUS
- COMPTON_WAVELENGTH
- CONDUCTANCE_QUANTUM
- CONVENTIONAL_VALUE_OF_AMPERE_90
- CONVENTIONAL_VALUE_OF_COULOMB_90
- CONVENTIONAL_VALUE_OF_FARAD_90
- CONVENTIONAL_VALUE_OF_HENRY_90
- CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT
- CONVENTIONAL_VALUE_OF_OHM_90
- CONVENTIONAL_VALUE_OF_VOLT_90
- CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT
- CONVENTIONAL_VALUE_OF_WATT_90
- COPPER_X_UNIT
- DEUTERON_ELECTRON_MAG_MOM_RATIO
- DEUTERON_ELECTRON_MASS_RATIO
- DEUTERON_G_FACTOR
- DEUTERON_MAG_MOM
- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- DEUTERON_MASS
- DEUTERON_MASS_ENERGY_EQUIVALENT

- DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV
- DEUTERON_MASS_IN_U
- DEUTERON_MOLAR_MASS
- DEUTERON_NEUTRON_MAG_MOM_RATIO
- DEUTERON_PROTON_MAG_MOM_RATIO
- DEUTERON_PROTON_MASS_RATIO
- DEUTERON_RELATIVE_ATOMIC_MASS
- DEUTERON_RMS_CHARGE_RADIUS
- ELECTRON_CHARGE_TO_MASS_QUOTIENT
- ELECTRON_DEUTERON_MAG_MOM_RATIO
- ELECTRON_DEUTERON_MASS_RATIO
- ELECTRON_G_FACTOR
- ELECTRON_GYROMAG_RATIO
- ELECTRON_GYROMAG_RATIO_IN_MHZ_T
- ELECTRON_HELION_MASS_RATIO
- ELECTRON_MAG_MOM
- ELECTRON_MAG_MOM_ANOMALY
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- ELECTRON_MASS
- ELECTRON_MASS_ENERGY_EQUIVALENT
- ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV
- ELECTRON_MASS_IN_U
- ELECTRON_MOLAR_MASS
- ELECTRON_MUON_MAG_MOM_RATIO
- ELECTRON_MUON_MASS_RATIO
- ELECTRON_NEUTRON_MAG_MOM_RATIO
- ELECTRON_NEUTRON_MASS_RATIO
- ELECTRON_PROTON_MAG_MOM_RATIO
- ELECTRON_PROTON_MASS_RATIO
- ELECTRON_RELATIVE_ATOMIC_MASS
- ELECTRON_TAU_MASS_RATIO
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO
- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO
- ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO
- ELECTRON_TRITON_MASS_RATIO
- ELECTRON_VOLT
- ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP

- ELECTRON_VOLT_HARTREE_RELATIONSHIP
- ELECTRON_VOLT_HERTZ_RELATIONSHIP
- ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP
- ELECTRON_VOLT_JOULE_RELATIONSHIP
- ELECTRON_VOLT_KELVIN_RELATIONSHIP
- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP
- ELEMENTARY_CHARGE
- ELEMENTARY_CHARGE_OVER_H_BAR
- FARADAY_CONSTANT
- FERMI_COUPLING_CONSTANT
- FINE_STRUCTURE_CONSTANT
- FIRST_RADIATION_CONSTANT
- FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE
- HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP
- HARTREE_ELECTRON_VOLT_RELATIONSHIP
- HARTREE_ENERGY
- HARTREE_ENERGY_IN_EV
- HARTREE_HERTZ_RELATIONSHIP
- HARTREE_INVERSE_METER_RELATIONSHIP
- HARTREE_JOULE_RELATIONSHIP
- HARTREE_KELVIN_RELATIONSHIP
- HARTREE_KILOGRAM_RELATIONSHIP
- HELION_ELECTRON_MASS_RATIO
- HELION_G_FACTOR
- HELION_MAG_MOM
- HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- HELION_MASS
- HELION_MASS_ENERGY_EQUIVALENT
- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV
- HELION_MASS_IN_U
- HELION_MOLAR_MASS
- HELION_PROTON_MASS_RATIO
- HELION_RELATIVE_ATOMIC_MASS
- HELION_SHIELDING_SHIFT
- HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP
- HERTZ_ELECTRON_VOLT_RELATIONSHIP
- HERTZ_HARTREE_RELATIONSHIP

- HERTZ_INVERSE_METER_RELATIONSHIP
- HERTZ_JOULE_RELATIONSHIP
- HERTZ_KELVIN_RELATIONSHIP
- HERTZ_KILOGRAM_RELATIONSHIP
- HYPERFINE_TRANSITION_FREQUENCY_OF_CS_133
- INVERSE_FINE_STRUCTURE_CONSTANT
- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP
- INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP
- INVERSE_METER_HARTREE_RELATIONSHIP
- INVERSE_METER_HERTZ_RELATIONSHIP
- INVERSE_METER_JOULE_RELATIONSHIP
- INVERSE_METER_KELVIN_RELATIONSHIP
- INVERSE_METER_KILOGRAM_RELATIONSHIP
- INVERSE_OF_CONDUCTANCE_QUANTUM
- JOSEPHSON_CONSTANT
- JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP
- JOULE_ELECTRON_VOLT_RELATIONSHIP
- JOULE_HARTREE_RELATIONSHIP
- JOULE_HERTZ_RELATIONSHIP
- JOULE_INVERSE_METER_RELATIONSHIP
- JOULE_KELVIN_RELATIONSHIP
- JOULE_KILOGRAM_RELATIONSHIP
- KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP
- KELVIN_ELECTRON_VOLT_RELATIONSHIP
- KELVIN_HARTREE_RELATIONSHIP
- KELVIN_HERTZ_RELATIONSHIP
- KELVIN_INVERSE_METER_RELATIONSHIP
- KELVIN_JOULE_RELATIONSHIP
- KELVIN_KILOGRAM_RELATIONSHIP
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP
- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP
- KILOGRAM_HARTREE_RELATIONSHIP
- KILOGRAM_HERTZ_RELATIONSHIP
- KILOGRAM_INVERSE_METER_RELATIONSHIP
- KILOGRAM_JOULE_RELATIONSHIP
- KILOGRAM_KELVIN_RELATIONSHIP
- LATTICE_PARAMETER_OF_SILICON
- LATTICE_SPACING_OF_IDEAL_SI_220

- LOSCHMIDT_CONSTANT_273_15_K_100_KPA
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA
- LUMINOUS_EFFICACY
- MAG_FLUX_QUANTUM
- MOLAR_GAS_CONSTANT
- MOLAR_MASS_CONSTANT
- MOLAR_MASS_OF_CARBON_12
- MOLAR_PLANCK_CONSTANT
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA
- MOLAR_VOLUME_OF_SILICON
- MOLYBDENUM_X_UNIT
- MUON_COMPTON_WAVELENGTH
- MUON_ELECTRON_MASS_RATIO
- MUON_G_FACTOR
- MUON_MAG_MOM
- MUON_MAG_MOM_ANOMALY
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- MUON_MASS
- MUON_MASS_ENERGY_EQUIVALENT
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV
- MUON_MASS_IN_U
- MUON_MOLAR_MASS
- MUON_NEUTRON_MASS_RATIO
- MUON_PROTON_MAG_MOM_RATIO
- MUON_PROTON_MASS_RATIO
- MUON_TAU_MASS_RATIO
- NATURAL_UNIT_OF_ACTION
- NATURAL_UNIT_OF_ACTION_IN_EV_S
- NATURAL_UNIT_OF_ENERGY
- NATURAL_UNIT_OF_ENERGY_IN_MEV
- NATURAL_UNIT_OF_LENGTH
- NATURAL_UNIT_OF_MASS
- NATURAL_UNIT_OF_MOMENTUM
- NATURAL_UNIT_OF_MOMENTUM_IN_MEV_C
- NATURAL_UNIT_OF_TIME
- NATURAL_UNIT_OF_VELOCITY

- NEUTRON_COMPTON_WAVELENGTH
- NEUTRON_ELECTRON_MAG_MOM_RATIO
- NEUTRON_ELECTRON_MASS_RATIO
- NEUTRON_G_FACTOR
- NEUTRON_GYROMAG_RATIO
- NEUTRON_GYROMAG_RATIO_IN_MHZ_T
- NEUTRON_MAG_MOM
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- NEUTRON_MASS
- NEUTRON_MASS_ENERGY_EQUIVALENT
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV
- NEUTRON_MASS_IN_U
- NEUTRON_MOLAR_MASS
- NEUTRON_MUON_MASS_RATIO
- NEUTRON_PROTON_MAG_MOM_RATIO
- NEUTRON_PROTON_MASS_DIFFERENCE
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U
- NEUTRON_PROTON_MASS_RATIO
- NEUTRON_RELATIVE_ATOMIC_MASS
- NEUTRON_TAU_MASS_RATIO
- NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO
- NEWTONIAN_CONSTANT_OF_GRAVITATION
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C
- NUCLEAR_MAGNETON
- NUCLEAR_MAGNETON_IN_EV_T
- NUCLEAR_MAGNETON_IN_INVERSE_METER_PER_TESLA
- NUCLEAR_MAGNETON_IN_K_T
- NUCLEAR_MAGNETON_IN_MHZ_T
- PLANCK_CONSTANT
- PLANCK_CONSTANT_IN_EV_HZ
- PLANCK_LENGTH
- PLANCK_MASS
- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV
- PLANCK_TEMPERATURE
- PLANCK_TIME

- PROTON_CHARGE_TO_MASS_QUOTIENT
- PROTON_COMPTON_WAVELENGTH
- PROTON_ELECTRON_MASS_RATIO
- PROTON_G_FACTOR
- PROTON_GYROMAG_RATIO
- PROTON_GYROMAG_RATIO_IN_MHZ_T
- PROTON_MAG_MOM
- PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- PROTON_MAG_SHIELDING_CORRECTION
- PROTON_MASS
- PROTON_MASS_ENERGY_EQUIVALENT
- PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV
- PROTON_MASS_IN_U
- PROTON_MOLAR_MASS
- PROTON_MUON_MASS_RATIO
- PROTON_NEUTRON_MAG_MOM_RATIO
- PROTON_NEUTRON_MASS_RATIO
- PROTON_RELATIVE_ATOMIC_MASS
- PROTON_RMS_CHARGE_RADIUS
- PROTON_TAU_MASS_RATIO
- QUANTUM_OF_CIRCULATION
- QUANTUM_OF_CIRCULATION_TIMES_2
- REDUCED_COMPTON_WAVELENGTH
- REDUCED_MUON_COMPTON_WAVELENGTH
- REDUCED_NEUTRON_COMPTON_WAVELENGTH
- REDUCED_PLANCK_CONSTANT
- REDUCED_PLANCK_CONSTANT_IN_EV_S
- REDUCED_PLANCK_CONSTANT_TIMES_C_IN_MEV_FM
- REDUCED_PROTON_COMPTON_WAVELENGTH
- REDUCED_TAU_COMPTON_WAVELENGTH
- RYDBERG_CONSTANT
- RYDBERG_CONSTANT_TIMES_C_IN_HZ
- RYDBERG_CONSTANT_TIMES_HC_IN_EV
- RYDBERG_CONSTANT_TIMES_HC_IN_J
- SACKUR_TETRODE_CONSTANT_1_K_100_KPA
- SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA
- SECOND_RADIATION_CONSTANT

- SHIELDED_HELION_GYROMAG_RATIO
- SHIELDED_HELION_GYROMAG_RATIO_IN_MHZ_T
- SHIELDED_HELION_MAG_MOM
- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO
- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO
- SHIELDED_PROTON_GYROMAG_RATIO
- SHIELDED_PROTON_GYROMAG_RATIO_IN_MHZ_T
- SHIELDED_PROTON_MAG_MOM
- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- SHIELDING_DIFFERENCE_OF_D_AND_P_IN_HD
- SHIELDING_DIFFERENCE_OF_T_AND_P_IN_HT
- SPEED_OF_LIGHT_IN_VACUUM
- STANDARD_ACCELERATION_OF_GRAVITY
- STANDARD_ATMOSPHERE
- STANDARD_STATE_PRESSURE
- STEFAN_BOLTZMANN_CONSTANT
- TAU_COMPTON_WAVELENGTH
- TAU_ELECTRON_MASS_RATIO
- TAU_ENERGY_EQUIVALENT
- TAU_MASS
- TAU_MASS_ENERGY_EQUIVALENT
- TAU_MASS_IN_U
- TAU_MOLAR_MASS
- TAU_MUON_MASS_RATIO
- TAU_NEUTRON_MASS_RATIO
- TAU_PROTON_MASS_RATIO
- THOMSON_CROSS_SECTION
- TRITON_ELECTRON_MASS_RATIO
- TRITON_G_FACTOR
- TRITON_MAG_MOM
- TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO
- TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO
- TRITON_MASS
- TRITON_MASS_ENERGY_EQUIVALENT
- TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV

- TRITON_MASS_IN_U
- TRITON_MOLAR_MASS
- TRITON_PROTON_MASS_RATIO
- TRITON_RELATIVE_ATOMIC_MASS
- TRITON_TO_PROTON_MAG_MOM_RATIO
- UNIFIED_ATOMIC_MASS_UNIT
- VACUUM_ELECTRIC_PERMITTIVITY
- VACUUM_MAG_PERMEABILITY
- VON_KLITZING_CONSTANT
- WEAK_MIXING_ANGLE
- WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT
- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT
- W_TO_Z_MASS_RATIO

CODATA 2018

List of available constants:

- ALPHA_PARTICLE_ELECTRON_MASS_RATIO _2018
- ALPHA_PARTICLE_MASS _2018
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT _2018
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV _2018
- ALPHA_PARTICLE_MASS_IN_U _2018
- ALPHA_PARTICLE_MOLAR_MASS _2018
- ALPHA_PARTICLE_PROTON_MASS_RATIO _2018
- ALPHA_PARTICLE_RELATIVE_ATOMIC_MASS _2018
- ANGSTROM_STAR _2018
- ATOMIC_MASS_CONSTANT _2018
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT _2018
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV _2018
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- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP_2010
- KILOGRAM_HARTREE_RELATIONSHIP_2010
- KILOGRAM_HERTZ_RELATIONSHIP_2010
- KILOGRAM_INVERSE_METER_RELATIONSHIP_2010
- KILOGRAM_JOULE_RELATIONSHIP_2010
- KILOGRAM_KELVIN_RELATIONSHIP_2010
- LATTICE_PARAMETER_OF_SILICON_2010
- LOSCHMIDT_CONSTANT_273_15_K_100_KPA_2010
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA_2010
- MAG_CONSTANT_2010
- MAG_FLUX_QUANTUM_2010
- MOLAR_GAS_CONSTANT_2010
- MOLAR_MASS_CONSTANT_2010
- MOLAR_MASS_OF_CARBON_12_2010
- MOLAR_PLANCK_CONSTANT_2010
- MOLAR_PLANCK_CONSTANT_TIMES_C_2010
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA_2010
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA_2010
- MOLAR_VOLUME_OF_SILICON_2010
- MO_X_UNIT_2010
- MUON_COMPTON_WAVELENGTH_2010
- MUON_COMPTON_WAVELENGTH_OVER_2_PI_2010
- MUON_ELECTRON_MASS_RATIO_2010
- MUON_G_FACTOR_2010
- MUON_MAG_MOM_2010
- MUON_MAG_MOM_ANOMALY_2010
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- MUON_MASS_2010
- MUON_MASS_ENERGY_EQUIVALENT_2010
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- MUON_MASS_IN_U_2010

- MUON_MOLAR_MASS_2010
- MUON_NEUTRON_MASS_RATIO_2010
- MUON_PROTON_MAG_MOM_RATIO_2010
- MUON_PROTON_MASS_RATIO_2010
- MUON_TAU_MASS_RATIO_2010
- NATURAL_UNIT_OF_ACTION_2010
- NATURAL_UNIT_OF_ACTION_IN_EV_S_2010
- NATURAL_UNIT_OF_ENERGY_2010
- NATURAL_UNIT_OF_ENERGY_IN_MEV_2010
- NATURAL_UNIT_OF_LENGTH_2010
- NATURAL_UNIT_OF_MASS_2010
- NATURAL_UNIT_OF_MOMUM_2010
- NATURAL_UNIT_OF_MOMUM_IN_MEV_C_2010
- NATURAL_UNIT_OF_TIME_2010
- NATURAL_UNIT_OF_VELOCITY_2010
- NEUTRON_COMPTON_WAVELENGTH_2010
- NEUTRON_COMPTON_WAVELENGTH_OVER_2_PI_2010
- NEUTRON_ELECTRON_MAG_MOM_RATIO_2010
- NEUTRON_ELECTRON_MASS_RATIO_2010
- NEUTRON_G_FACTOR_2010
- NEUTRON_GYROMAG_RATIO_2010
- NEUTRON_GYROMAG_RATIO_OVER_2_PI_2010
- NEUTRON_MAG_MOM_2010
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- NEUTRON_MASS_2010
- NEUTRON_MASS_ENERGY_EQUIVALENT_2010
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- NEUTRON_MASS_IN_U_2010
- NEUTRON_MOLAR_MASS_2010
- NEUTRON_MUON_MASS_RATIO_2010
- NEUTRON_PROTON_MAG_MOM_RATIO_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV_2010
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U_2010
- NEUTRON_PROTON_MASS_RATIO_2010
- NEUTRON_TAU_MASS_RATIO_2010

- NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO _2010
- NEWTONIAN_CONSTANT_OF_GRAVITATION _2010
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C _2010
- NUCLEAR_MAGNETON _2010
- NUCLEAR_MAGNETON_IN_EV_T _2010
- NUCLEAR_MAGNETON_IN_INVERSE_METERS_PER_TESLA _2010
- NUCLEAR_MAGNETON_IN_K_T _2010
- NUCLEAR_MAGNETON_IN_MHZ_T _2010
- PLANCK_CONSTANT _2010
- PLANCK_CONSTANT_IN_EV_S _2010
- PLANCK_CONSTANT_OVER_2_PI _2010
- PLANCK_CONSTANT_OVER_2_PI_IN_EV_S _2010
- PLANCK_CONSTANT_OVER_2_PI_TIMES_C_IN_MEV_FM _2010
- PLANCK_LENGTH _2010
- PLANCK_MASS _2010
- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV _2010
- PLANCK_TEMPERATURE _2010
- PLANCK_TIME _2010
- PROTON_CHARGE_TO_MASS_QUOTIENT _2010
- PROTON_COMPTON_WAVELENGTH _2010
- PROTON_COMPTON_WAVELENGTH_OVER_2_PI _2010
- PROTON_ELECTRON_MASS_RATIO _2010
- PROTON_G_FACTOR _2010
- PROTON_GYROMAG_RATIO _2010
- PROTON_GYROMAG_RATIO_OVER_2_PI _2010
- PROTON_MAG_MOM _2010
- PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO _2010
- PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO _2010
- PROTON_MAG_SHIELDING_CORRECTION _2010
- PROTON_MASS _2010
- PROTON_MASS_ENERGY_EQUIVALENT _2010
- PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV _2010
- PROTON_MASS_IN_U _2010
- PROTON_MOLAR_MASS _2010
- PROTON_MUON_MASS_RATIO _2010
- PROTON_NEUTRON_MAG_MOM_RATIO _2010
- PROTON_NEUTRON_MASS_RATIO _2010
- PROTON_RMS_CHARGE_RADIUS _2010

- PROTON_TAU_MASS_RATIO_2010
- QUANTUM_OF_CIRCULATION_2010
- QUANTUM_OF_CIRCULATION_TIMES_2_2010
- RYDBERG_CONSTANT_2010
- RYDBERG_CONSTANT_TIMES_C_IN_HZ_2010
- RYDBERG_CONSTANT_TIMES_HC_IN_EV_2010
- RYDBERG_CONSTANT_TIMES_HC_IN_J_2010
- SACKUR_TETRODE_CONSTANT_1_K_100_KPA_2010
- SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA_2010
- SECOND_RADIATION_CONSTANT_2010
- SHIELDED_HELION_GYROMAG_RATIO_2010
- SHIELDED_HELION_GYROMAG_RATIO_OVER_2_PI_2010
- SHIELDED_HELION_MAG_MOM_2010
- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO_2010
- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2010
- SHIELDED_PROTON_GYROMAG_RATIO_2010
- SHIELDED_PROTON_GYROMAG_RATIO_OVER_2_PI_2010
- SHIELDED_PROTON_MAG_MOM_2010
- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- SPEED_OF_LIGHT_IN_VACUUM_2010
- STANDARD_ACCELERATION_OF_GRAVITY_2010
- STANDARD_ATMOSPHERE_2010
- STANDARD_STATE_PRESSURE_2010
- STEFAN_BOLTZMANN_CONSTANT_2010
- TAU_COMPTON_WAVELENGTH_2010
- TAU_COMPTON_WAVELENGTH_OVER_2_PI_2010
- TAU_ELECTRON_MASS_RATIO_2010
- TAU_MASS_2010
- TAU_MASS_ENERGY_EQUIVALENT_2010
- TAU_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- TAU_MASS_IN_U_2010
- TAU_MOLAR_MASS_2010
- TAU_MUON_MASS_RATIO_2010
- TAU_NEUTRON_MASS_RATIO_2010
- TAU_PROTON_MASS_RATIO_2010

- THOMSON_CROSS_SECTION_2010
- TRITON_ELECTRON_MASS_RATIO_2010
- TRITON_G_FACTOR_2010
- TRITON_MAG_MOM_2010
- TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2010
- TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2010
- TRITON_MASS_2010
- TRITON_MASS_ENERGY_EQUIVALENT_2010
- TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV_2010
- TRITON_MASS_IN_U_2010
- TRITON_MOLAR_MASS_2010
- TRITON_PROTON_MASS_RATIO_2010
- UNIFIED_ATOMIC_MASS_UNIT_2010
- VON_KLITZING_CONSTANT_2010
- WEAK_MIXING_ANGLE_2010
- WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT_2010
- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT_2010

NAME

codata - print fundamental physical constants

SYNOPSIS

codata [OPTION...] [REGEX_PATTERN...]

DESCRIPTION

codata(1) is a command line interface which prints all the **codata** constants.

The current values are from 2022. Older values can be retrieved if needed and the output can be filtered with REGEX PATTERNS.

OPTIONS

--year, -y YEAR

CODATA constants: 2022, 2018, 2014, 2010

--pattern, -p PATTERN

Regex pattern for filtering the constants.

--value, -a

Show only the value.

--error, -e

Show only the uncertainty.

--usage Show usage text and exit.

--version, -v

Show version information and exit.

--help, -h

Show help text and exit.

NOTES

You may replace the default options from a file if your first options begin with @file. Initial options will then be read from the "response file" "file.rsp" in the current directory.

If "file" does not exist or cannot be read, then an error occurs and the program stops. Each line of the file is prefixed with "options" and interpreted as a separate argument. The file itself may not contain @file arguments. That is, it is not processed recursively.

For more information on response files see https://urbanjost.github.io/M_CLI2/set_args.3m_cli2.html

EXAMPLE

Minimal example

```
codata
codata -y 2018 molar electron
codata -y 2014 -p molar.*gas,electron.*eV
codata [B,b]oltzmann.*eV
```

SEE ALSO

codata(3)

NAME

version - get the version of the Fortran library codata

LIBRARY

codata (**-libcodata**, **-lcodata**)

SYNOPSIS

Fortran: **character**(*len=:*), pointer **version**()

C: char* **codata_version**()

Python: pycodata.__version__

DESCRIPTION

Get the library version.

EXAMPLE

Fortran:

```
use codata
print *, "version = ", version()
```

C:

```
include <stdio.h>
include "codata.h"
printf("version = %s0, codata_version());
```

Python:

```
import pycodata
print(f"version = {pycodata.__version__}")
```

SEE ALSO

codata(3)

NAME

codata_constant_type - data type defining a constant

SYNOPSIS

Fortran: **type**(*codata_constant_type*) :: c

C: struct *codata_constant_type* c

DESCRIPTION

Defines a constant with the name, the value, the uncertainty and the unit.

EXAMPLE

Fortran:

```
use codata
type(codata_constant_type), pointer :: c
c = SPEED_OF_LIGHT_IN_VACUUM
```

C:

```
struct codata_constant_type c;
c = SPEED_OF_LIGHT_IN_VACUUM;
```

SEE ALSO

codata(3)